12TH BIOLOGY · CHAPTER 24 · FULL CHAPTER MCQS

Evolution

52 topic-wise MCQs from the complete chapter with answers and reasons

24.1 – 24.2 CONCEPT OF EVOLUTION AND DARWINISM

1. The theory that all organisms were created as they exist today is:

(a) Lamarckism
(b) Endosymbiosis
(c) Darwinism
(d) Creationism

Answer: (d) Creationism — Also called the theory of intelligent design.

2. The idea that living creatures develop from non-living matter is:

(a) Natural selection
(b) Spontaneous generation
(c) Gene flow
(d) Homology

Answer: (b) Spontaneous generation — Experiments disproved it.

3. Evolution is the gradual change in a population's:

(a) Genetic makeup (allele frequencies)
(b) Body size
(c) Habitat
(d) Food

Answer: (a) Genetic makeup (allele frequencies) — Over successive generations.

4. Darwin sailed on HMS Beagle in:

(a) 1809
(b) 1859
(c) 1831
(d) 1908

Answer: (c) 1831 — The voyage lasted 5 years.

5. The Galápagos Islands are a group of how many small islands?

(a) 16
(b) 6
(c) 26
(d) 60

Answer: (a) 16 — As given in the textbook.

6. Giant tortoises on different Galápagos islands differed in:

(a) Colour of eyes
(b) Shape of shell
(c) Number of legs
(d) Type of blood

Answer: (b) Shape of shell — Saddle-shaped or dome-shaped.

7. Beaks of Galápagos finches differed as adaptations for:

(a) Flight
(b) Singing
(c) Different kinds of food
(d) Nest building

Answer: (c) Different kinds of food — On different islands.

studysapiens.com

8. "On the Origin of Species" was published in:

(a) 1831 (b) 1858
(c) 1809 (d) 1859

Answer: (d) 1859 — By Charles Darwin.

9. A female sea star releasing about 1 million eggs illustrates:

(a) Overproduction (b) Variation
(c) Gene flow (d) Homology

Answer: (a) Overproduction — Reproductive potential is far greater than realized.

10. Advantageous variations that help reproduction are called:

(a) Acquired characters (b) Adaptive traits
(c) Vestigial traits (d) Neutral traits

Answer: (b) Adaptive traits — They are inherited.

11. Darwin's and Wallace's theories were presented in 1858 to the:

(a) Royal Society (b) Zoological Society
(c) French Academy (d) Linnaean Society

Answer: (d) Linnaean Society — Wallace attributed the theory to Darwin.

12. Neo-Darwinism integrated Darwinism with:

(a) Ecology (b) Anatomy
(c) Modern genetics (d) Geology

Answer: (c) Modern genetics — In the early 20th century.

13. The total number of all genes for a trait in a population is its:

(a) Genotype (b) Gene pool
(c) Genome size (d) Karyotype

Answer: (b) Gene pool — Neo-Darwinism defines evolution by it.

14. If 600 of 1000 genes are for red colour, the allele frequency of red is:

(a) 0.6 (b) 0.4
(c) 6 (d) 0.06

Answer: (a) 0.6 — 600 ÷ 1000.

15. The Hardy-Weinberg theorem was presented in:

(a) 1859 (b) 1809
(c) 1908 (d) 1932

Answer: (c) 1908 — By Hardy and Weinberg.

16. Wilhelm Weinberg was a German:

(a) Mathematician (b) Geologist
(c) Botanist (d) Physician

Answer: (d) Physician — Hardy was an English mathematician.

17. Hardy-Weinberg equilibrium requires a population that is:

(a) Small (b) Large
(c) Migrating (d) Mutating

Answer: (b) Large — With random mating.

18. Genetic drift is most obvious in:

(a) Small populations (b) Large populations
(c) Plants only (d) Bacteria only

Answer: (a) Small populations — Chance has a greater effect there.

19. Genetic drift is also called:

(a) Natural selection (b) Gene flow
(c) Neutral selection (d) Artificial selection

Answer: (c) Neutral selection — It works by chance, not fitness.

20. The origin of all new genes is:

(a) Gene flow (b) Migration
(c) Genetic drift (d) Mutation

Answer: (d) Mutation — It is a source of variation.

21. Most mutations are:

(a) Beneficial (b) Neutral
(c) Deleterious (d) Inherited acquired characters

Answer: (c) Deleterious — As stated in the textbook.

22. The tendency of gene frequencies to change through mutation is:

(a) Mutational equilibrium (b) Mutation pressure
(c) Gene flow (d) Drift

Answer: (b) Mutation pressure — It is a measure.

24.3 EVIDENCES OF EVOLUTION

23. The study of the geographic distribution of plants and animals is:

(a) Biogeography (b) Biochemistry
(c) Anatomy (d) Palaeontology

Answer: (a) Biogeography — It gives evidence of evolution.

24. The African lion is very similar to the:

(a) Asian tiger (b) Kangaroo
(c) Mountain lion of America (d) Leopard

Answer: (c) Mountain lion of America — This suggests a common ancestry.

studysapiens.com

25. The major herbivores of Australia and Tasmania are:

(a) Deer (b) Kangaroos
(c) Horses (d) Giraffes

Answer: (b) Kangaroos — They differ from animals elsewhere.

26. Islands often have species that are:

(a) Many and common (b) All extinct
(c) Absent (d) Few but unique

Answer: (d) Few but unique — Due to island colonization.

27. Structures similar in structure but different in function are:

(a) Homologous (b) Analogous
(c) Vestigial (d) Acquired

Answer: (a) Homologous — They show divergent evolution.

28. Homologous structures reveal:

(a) Convergent evolution (b) Spontaneous generation
(c) Genetic drift (d) Divergent evolution

Answer: (d) Divergent evolution — A common ancestor.

29. The wing of a bird and the wing of an insect are:

(a) Homologous (b) Vestigial
(c) Analogous (d) Identical

Answer: (c) Analogous — Same function, different structure.

30. Analogous organs indicate:

(a) Divergent evolution (b) Convergent evolution
(c) Common ancestry (d) Gene flow

Answer: (b) Convergent evolution — Different ancestors in a common habitat.

31. Minute remnants of hindlimb bones are found in:

(a) Frogs (b) Whales only
(c) Lizards (d) Pythons

Answer: (d) Pythons — Also in boa constrictors.

32. The number of vestigial structures in the human body is about:

(a) 9 (b) 90
(c) 19 (d) 900

Answer: (b) 90 — As given in the textbook.

33. The vermiform appendix is a vestige of the:

(a) Cecum (b) Colon
(c) Stomach (d) Rectum

Answer: (a) Cecum — A human vestigial structure.

studysapiens.com

34. The coccyx is the remnant of the:

(a) Hindlimb (b) Rib
(c) Tail (d) Jaw

Answer: (c) Tail — It is the tailbone.

35. Wisdom teeth are vestigial:

(a) Incisors (b) Premolars
(c) Canines (d) Third molars

Answer: (d) Third molars — Once used for grinding plant tissue.

36. Related animals have similar proteins because they have similar:

(a) DNA (b) Habitats
(c) Food (d) Body size

Answer: (a) DNA — DNA codes for proteins.

37. Antibiotic resistance in bacteria may arise due to:

(a) Use and disuse (b) Gene therapy
(c) Mutations in bacterial genes (d) Vestigial organs

Answer: (c) Mutations in bacterial genes — It then spreads by natural selection.

24.4 EVOLUTION FROM PROKARYOTES TO EUKARYOTES

38. Prokaryotes originated about:

(a) 2 billion years ago (b) 1 million years ago
(c) 3.5 billion years ago (d) 500 million years ago

Answer: (c) 3.5 billion years ago — They were the earliest organisms.

39. Eukaryotes evolved from prokaryotes about:

(a) 2 billion years ago (b) 3.5 billion years ago
(c) 5 billion years ago (d) 1 billion years ago

Answer: (a) 2 billion years ago — As given in the textbook.

40. The endosymbiotic theory was proposed by:

(a) Darwin (b) Lynn Margulis
(c) Lamarck (d) Wallace

Answer: (b) Lynn Margulis — The most acceptable theory.

41. Small prokaryotes capable of aerobic respiration became:

(a) Chloroplasts (b) Ribosomes
(c) Nuclei (d) Mitochondria

Answer: (d) Mitochondria — They supplied energy.

studysapiens.com

42. Small prokaryotes that used sunlight to make food became:

(a) Mitochondria (b) Flagella
(c) Chloroplasts (d) Lysosomes

Answer: (c) Chloroplasts — They shared food with the large cell.

43. Flagella may have come from prokaryotes similar to:

(a) Spirochetes (b) Cyanobacteria
(c) Cocci (d) Viruses

Answer: (a) Spirochetes — By ingestion.

44. Mitochondria reproduce by:

(a) Mitosis (b) Budding
(c) Meiosis (d) Binary fission

Answer: (d) Binary fission — Like prokaryotes.

45. The DNA of mitochondria is organized as a:

(a) Linear chromosome (b) Circular chromosome
(c) Double nucleus (d) Plasmid only

Answer: (b) Circular chromosome — Like a prokaryotic chromosome.

46. Origin of the nucleus and endoplasmic reticulum is explained by the:

(a) Endosymbiotic theory (b) Hardy-Weinberg theorem
(c) Lamarckism (d) Membrane invagination theory

Answer: (d) Membrane invagination theory — By infolding of the cell membrane.

24.5 LAMARCKISM

47. Lamarck was a naturalist from:

(a) England (b) Italy
(c) Germany (d) France

Answer: (d) France — He lived from 1744 to 1829.

48. Lamarck published “Philosophie Zoologique” in:

(a) 1859 (b) 1831
(c) 1809 (d) 1908

Answer: (c) 1809 — It presented Lamarckism.

49. Changes in body parts due to long-term use or disuse are:

(a) Acquired characters (b) Adaptive traits
(c) Mutations (d) Vestigial organs

Answer: (a) Acquired characters — Lamarck’s term.

studysapiens.com

50. The usual example given to explain Lamarckism is the:

(a) Beak of finch (b) Neck of giraffe
(c) Shell of tortoise (d) Wing of bat

Answer: (b) Neck of giraffe — It became long by stretching.

51. Lamarckism was disproved because acquired characters:

(a) Have no genetic basis (b) Are harmful
(c) Are too many (d) Occur in plants only

Answer: (a) Have no genetic basis — So they cannot be inherited.

52. According to Lamarck, organs that are not used:

(a) Become strong (b) Mutate
(c) Deteriorate or become weak (d) Multiply

Answer: (c) Deteriorate or become weak — Disuse of organs.

studysapiens.com